

MARIE CATHERINE DUVAL

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PROFESSIONAL SUMMARY

Results-driven Data Scientist with 3 years of experience building machine learning models and driving data-driven decision-making across diverse industries. Skilled in statistical analysis, predictive modeling, and translating complex data insights into actionable business strategies. Passionate about leveraging advanced analytics to solve real-world problems and optimizing model performance. Seeking to advance my career by tackling more complex data challenges in a dynamic analytical environment.

WORK EXPERIENCE

Data Scientist

DataFlow Innovations | <https://www.dataflow-innovations.io>

Paris, France

January 2023 - Present

Developed and deployed 5 machine learning models for customer churn prediction and retention optimization, achieving 87% accuracy and reducing customer attrition by 12% within first quarter

Built automated ETL pipelines using Python and Apache Spark to process 50GB+ monthly datasets from multiple data sources, reducing manual data preparation time by 60%

Conducted A/B testing and statistical analysis on marketing campaigns, providing actionable insights that increased conversion rates by 9%

Collaborated with cross-functional teams including product and marketing to translate business requirements into analytical solutions and documented findings in comprehensive reports

Senior Data Analyst

Nexus Analytics Solutions | <https://www.nexus-analytics-solutions.net>

Lyon, France

June 2021 - December 2022

Previously known as Vertex Intelligence Consulting

Engineered predictive models using scikit-learn and XGBoost for demand forecasting, improving inventory optimization accuracy by 18%

Managed end-to-end data pipeline architecture and implemented data quality frameworks that reduced reporting errors by 35%

Created interactive dashboards using Tableau and Python visualization libraries, enabling stakeholders to monitor KPIs in real-time

Mentored 2 junior analysts on best practices in statistical modeling and SQL query optimization, fostering a culture of continuous learning

Data Scientist

Société Générale | <https://www.societegenerale.com>

Paris, France

August 2020 - May 2021

Analyzed large-scale transactional datasets to identify fraud patterns and developed classification models achieving 94% precision in anomaly detection

Performed exploratory data analysis and feature engineering on customer behavioral data to support credit risk assessment models

Collaborated with IT and security teams to implement automated monitoring systems for model performance degradation and data drift

Presented quarterly findings to senior management and provided recommendations for model refinement and strategic adjustments

EDUCATION

Master's Degree in Data Science and Machine Learning

Université Paris-Saclay, Paris, France

Graduated: June 2020

Bachelor's Degree in Statistics and Mathematics

Université Pierre et Marie Curie, Paris, France

Graduated: June 2018

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Machine Learning Frameworks: scikit-learn, TensorFlow, XGBoost, LightGBM

Data Processing: Pandas, NumPy, Apache Spark, PySpark

Database Management: PostgreSQL, MySQL, MongoDB

Data Visualization: Tableau, Matplotlib, Seaborn, Plotly

Statistical Methods: Hypothesis Testing, Regression Analysis, Time Series Analysis, Clustering

Cloud Platforms: AWS, Google Cloud Platform

Tools and Platforms: Jupyter Notebook, Git, Unix/Linux, Jira

Domain Knowledge: Customer Analytics, Fraud Detection, Predictive Modeling, Business Intelligence

CERTIFICATIONS

AWS Certified Cloud Practitioner

Completed: March 2022

Google Analytics Professional Certificate

Completed: September 2021

LANGUAGES

French (Native)

English (Fluent)

Spanish (Conversational)